A First Course in Abstract Algebra

Section 19: Integral Domains

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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19 Integral Domains

Exercise 19.1: A cubic congruence

Factor $x^3 - 2x^2 - 3x = x(x-3)(x+1)$. Because $12 = 3 \cdot 4$, it vanishes modulo 12 exactly when it vanishes modulo both 3 and 4. Modulo 3 the admissible residues are $0, 2$; modulo 4 they are $0, 1, 3$. Pairing these conditions, or testing the twelve residues, gives

$$\boxed{x = 0, 3, 5, 8, 9, 11 \text{ in } \mathbb{Z}_{12}}.$$

For example $x = 8$ gives $8 \cdot 5 \cdot 9 = 360$, divisible by 12. Factoring alone does not permit setting one factor equal to zero in a ring with zero divisors.

Exercise 19.2: Solving linear equations in two prime fields

In $\mathbb{Z}_7$, $3^{-1} = 5$, since $3 \cdot 5 = 15 \equiv 1$; hence $x = 5 \cdot 2 = \boxed{3}$. In $\mathbb{Z}_{23}$, $3^{-1} = 8$, since $3 \cdot 8 = 24 \equiv 1$; hence $x = \boxed{16}$. Multiplication by a nonzero field element is bijective, so these solutions are unique.

Exercise 19.3: A quadratic modulo six

For $x = 0, 1, 2, 3, 4, 5$, the residues of $x^2 + 2x + 2$ modulo 6 are respectively $2, 5, 4, 5, 2, 1$. None is zero, so there are $\boxed{\text{no solutions}}$. This exhaustive check is valid because these are all six elements of the ring.

Exercise 19.4: A second quadratic modulo six

Evaluating $x^2 + 2x + 4$ at $x = 0, 1, 2, 3, 4, 5$ gives residues $4, 1, 0, 1, 4, 3$ modulo 6. Therefore the complete solution set is $\boxed{\{2\}}$.

Exercise 19.5: Characteristic of the even integers

The characteristic is 0. For every positive integer n, the sum of n copies of the element 2 is the nonzero integer $2n$. Thus no positive n annihilates every element.

Exercise 19.6: Characteristic of two integer coordinates

For every positive n, $n(1, 0) = (n, 0) \neq (0, 0)$. Hence there is no common positive additive annihilator, and $\boxed{\text{char}(\mathbb{Z} \times \mathbb{Z}) = 0}$.

Exercise 19.7: A finite and an infinite coordinate

Although the first factor $\mathbb{Z}_3$ has characteristic 3, the element $(0, 3)$ in $\mathbb{Z}_3 \times 3\mathbb{Z}$ has infinite additive order. Its n-fold sum is $(0, 3n) \neq 0$ for every positive n. The product therefore has characteristic $\boxed{0}$.

Exercise 19.8: Two characteristic-three factors

Three copies of any element of $\mathbb{Z}_3 \times \mathbb{Z}_3$ sum to zero. One or two copies of $(1,0)$ do not. The least common positive annihilator is consequently $\boxed{3}$.

Exercise 19.9: Relatively prime finite characteristics

A positive integer annihilates $\mathbb{Z}_3 \times \mathbb{Z}_4$ exactly when it is divisible by both 3 and 4. The least such integer is $\operatorname{lcm}(3,4) = \boxed{12}$.

Exercise 19.10: A least common multiple of characteristics

The characteristic of $\mathbb{Z}_6 \times \mathbb{Z}_{15}$ is $\operatorname{lcm}(6,15) = \boxed{30}$. Indeed $n(1,0) = 0$ requires $6 \mid n$, while $n(0,1) = 0$ requires $15 \mid n$; these two conditions are also sufficient to annihilate every element.

Exercise 19.11: A fourth power in characteristic four

Commutativity allows the binomial theorem:

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

In characteristic 4, coefficients are reduced modulo 4: $4 = 0$ and $6 = 2$. Thus the simplified expression is $\boxed{a^4 + 2a^2b^2 + b^4}$. The middle term need not vanish.

Exercise 19.12: A fifth power in characteristic three

The binomial coefficients for exponent 5 are $1, 5, 10, 10, 5, 1$, reducing modulo 3 to $1, 2, 1, 1, 2, 1$. Therefore

$$\boxed{(a+b)^5 = a^5 + 2a^4b + a^3b^2 + a^2b^3 + 2ab^4 + b^5}.$$

Each reduction is justified because $3r = 0$ for every element r of the ring.

Exercise 19.13: A sixth power in characteristic three

First $(a+b)^3 = a^3 + b^3$, since the intermediate binomial coefficients are 3. Squaring and using commutativity gives

$$\boxed{(a+b)^6 = (a^3+b^3)^2 = a^6 + 2a^3b^3 + b^6}.$$

This agrees with reducing the sixth-row binomial coefficients modulo 3.

Exercise 19.14: Exhibiting a matrix zero divisor

Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both matrices are nonzero, but $AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, as row-by-column multiplication verifies. Thus A is a divisor of zero in $M_2(\mathbb{Z})$.

Exercise 19.15: Correcting zero-divisor terminology

One must exclude zero factors. A nonzero element a is a divisor of zero if there is a nonzero element b with $ab = 0$ or $ba = 0$ (in a commutative ring these are the same condition). The mere equation $ab = 0$ does not suffice if, for instance, $b = 0$.

Exercise 19.16: Correcting characteristic

The characteristic is the *least positive integer* n such that $na = 0$ for every $a \in R$, if such a positive integer exists. If none exists, the characteristic is 0. The printed condition omits both minimality and the characteristic-zero case.

Exercise 19.17: True or false: integral domains

(a) False: $n\mathbb{Z}$, as a subring of $\mathbb{Z}$, never has nonzero zero divisors, regardless of primality.

(b) True: a field is commutative with nonzero unity, and $ab = 0$, $a \neq 0$ implies $b = a^{-1}0 = 0$.

(c) False: for positive n, $n\mathbb{Z}$ has characteristic 0, since $kn \neq 0$ for positive k.

(d) False: for $n > 1$, $n\mathbb{Z}$ has no unity, whereas $\mathbb{Z}$ has one; an onto isomorphism would transport unity. Equivalently compare the images of 1 and 1^2.

(e) True: an isomorphism transports a zero-product or a cancellation equation to the integral domain, where cancellation holds.

(f) True: the elements $1, 2 \cdot 1, 3 \cdot 1, \ldots$ are distinct in characteristic zero.

(g) False: in a product of two domains, $(1, 0)(0, 1) = 0$ with both factors nonzero.

(h) True: multiplying $ab = 0$ by the inverse of a would force $b = 0$.

(i) False in general: for $n > 1$, $n\mathbb{Z}$ lacks unity and is not an integral domain under the book's definition. For $n = 1$ it is a subdomain.

(j) False: $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it lacks the inverse of 2 and is not a field.

Exercise 19.18: Examples in the six ring regions

The regions of Figure 19.10 have the following examples.

1. $\mathbb{Q}$: a field.

2. $\mathbb{Z}$: an integral domain that is not a field, because 2 is not a unit.

3. $\mathbb{Z}_6$: a commutative ring with unity but with $2 \cdot 3 = 0$.

4. $2\mathbb{Z}$: a commutative ring without unity.

5. $M_2(\mathbb{R})$: a ring with unity but not commutative, since $E_{12}E_{21} = E_{11} \neq E_{22} = E_{21}E_{12}$.

6. $M_2(2\mathbb{Z})$: noncommutative by the same calculation using $2E_{12}, 2E_{21}$, and without unity. A unity would have to act as identity on $2E_{11}$ and $2E_{22}$, forcing it to be I_2, whose entries are not all even.

Exercise 19.19: Five matrix criteria for being a zero divisor

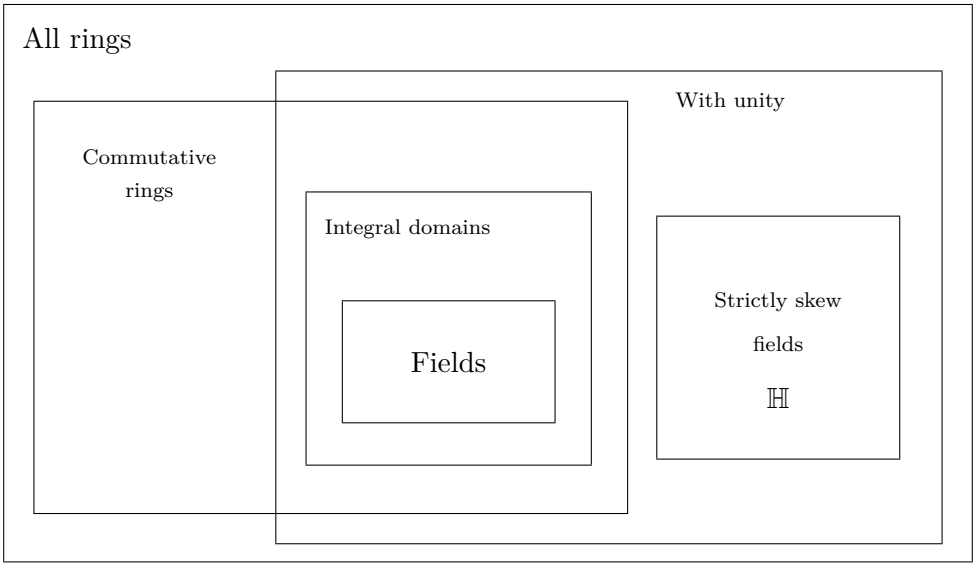
For a *nonzero* $A \in M_n(F)$, the following five equivalent tests characterize zero divisors:

$$\det A = 0; \quad \operatorname{rank} A < n; \quad \operatorname{Ker} A \neq \{0\}; \quad \text{columns of } A \text{ are dependent}; \quad A \text{ is not invertible.}$$

These are equivalent by the invertible-matrix theorem over a field. To connect them to the ring definition, if $Av = 0$ with $v \neq 0$, form B with first column v and all other columns zero. Then $B \neq 0$ and $AB = 0$. Conversely if $AB = 0$ with $B \neq 0$, some nonzero column v of B satisfies $Av = 0$. If instead $BA = 0$, invertibility would still be impossible. The exclusion $A \neq 0$ is essential because the book does not call zero itself a zero divisor.

Exercise 19.20: Locating strictly skew fields in the ring diagram

A strictly skew field is a division ring whose multiplication is not commutative. It therefore lies inside the rings-with-unity region and outside the commutative-rings region. Its set is disjoint from the integral domains and fields of the book's definitions. The quaternion division ring $\mathbb{H}$ supplies an example. The following inclusion diagram redraws the relevant regions using rectangles instead of circles.



Exercise 19.21: Synopsis of cancellation without zero divisors

If $ab = ac$ with $a \neq 0$, then $a(b - c) = 0$, so the absence of zero divisors forces $b - c = 0$ and hence $b = c$; the right-cancellation argument is identical with the factors reversed.

Exercise 19.22: Synopsis of the finite-domain theorem

For nonzero a in a finite integral domain, the map $x \mapsto ax$ is injective by cancellation and therefore surjective, so $ax = 1$ for some x, giving the required multiplicative inverse.

Exercise 19.23: Idempotents in a division ring

Both 0 and 1 are idempotent and are distinct. If $a^2 = a$ and $a \neq 0$, left multiplication by a^{-1} gives $a = 1$. Thus the only possibilities are exactly $\boxed{0, 1}$.

Exercise 19.24: Intersection of subdomains

A subdomain $D' \subseteq D$ must have the same unity as D: its nonzero unity e satisfies $e(e - 1_D) = 0$, hence $e = 1_D$ because D has no zero divisors. Thus an intersection of subdomains contains $1_D \neq 0$. It is a subring by the subring test, commutative by inheritance, and has no zero divisors because D has none. It is consequently a subdomain. For the empty family use the ambient domain as its intersection.

Exercise 19.25: A finite ring without zero divisors

Fix $a \neq 0$. The maps $x \mapsto ax$ and $x \mapsto xa$ are injective: equality of outputs gives a zero product after subtraction, so cancellation applies. Finiteness makes them onto. Hence there exist b, c with $ab = 1$ and $ca = 1$. They agree because $c = c(ab) = (ca)b = b$. Thus a has a two-sided inverse. This holds for every nonzero a, proving that R is a division ring; no commutativity was assumed.

Exercise 19.26: A unique generalized inverse forces a division ring

(a) Fix nonzero a and its unique b with $aba = a$. If either $ax = 0$ or $xa = 0$, then $axa = 0$, so $a(b + x)a = a$. Uniqueness implies $b + x = b$, hence $x = 0$. Thus there are no nonzero zero divisors.

(b) Associativity gives $a(bab)a = (aba)ba = aba = a$. Uniqueness of b gives $bab = b$.

(c) The element $e = ab$ is nonzero, since $aba = a \neq 0$, and $e^2 = (aba)b = ab = e$. For every x, $e(ex - x) = 0$ and $(xe - x)e = 0$. By (a), both differences are zero. Thus $ex = xe = x$, so e is a unity for the whole ring.

(d) Also ba is a nonzero idempotent and hence, by the same argument, is a unity. The uniqueness of unity gives $ba = ab = 1$. Every nonzero a therefore has a two-sided inverse, so R is a division ring.

Exercise 19.27: Characteristic of a subdomain

A subdomain $E \subseteq D$ has unity 1_D, as proved in Exercise 19.24. In any ring with unity the characteristic is the additive order of its unity, with infinite order recorded as characteristic zero. The sums $n1$ are identical whether computed in E or D, so their first positive zero, if any, occurs at the same n. Hence the two characteristics agree.

Exercise 19.28: The smallest subdomain

Let $P = \{n1_D : n \in \mathbb{Z}\}$. It contains 0 and 1_D, and

$$(m1_D) - (n1_D) = (m-n)1_D, \qquad (m1_D)(n1_D) = mn1_D.$$

It is therefore a commutative subring with nonzero unity, and has no zero divisors because it is contained in D. Thus it is a subdomain. Every subdomain contains the same unity 1_D and its integer sums and negatives, so every subdomain contains P.

Exercise 19.29: The characteristic of an integral domain

Suppose the positive characteristic is composite, say $n = rs$ with $1 < r, s < n$. Minimality gives $r1 \neq 0$ and $s1 \neq 0$, but

$$(r1)(s1) = rs1 = n1 = 0,$$

contradicting the absence of zero divisors. Also the characteristic cannot be 1, since $1 \neq 0$. Therefore the characteristic is either 0 or a prime.

Exercise 19.30: Adjoining a unity without changing characteristic

Write $K = \mathbb{Z}$ in characteristic zero, and $K = \mathbb{Z}_n$ in characteristic $n > 0$. On $S = R \times K$ use the stated operations. Integer scalar multiplication on R is well defined for residues modulo n because $nr = 0$.

(a) Addition is the direct-product abelian group. To check multiplication associativity, write three elements as $(r,k),(s,l),(t,m)$. The first coordinate of either associated product expands to

$$rst + k(st) + l(rt) + m(rs) + (kl)t + (km)s + (lm)r,$$

and the second coordinate is klm. The two distributive laws follow by expanding in the same way using distributivity in R and integer scalar multiplication. Thus this is a ring.

(b) Its unity is $(0,1)$, since

$$(0,1)(r,k) = (r,k) = (r,k)(0,1).$$

(c) If R has characteristic $n > 0$, both coordinates are killed by n, and the element $(0,1)$ has additive order exactly n. If R has characteristic zero, the integer coordinate of $(0,1)$ has infinite order. Therefore the characteristics agree. This includes the zero ring, with characteristic 1, if admitted.

(d) The map $r \mapsto (r,0)$ is injective and satisfies $(r,0)+(s,0) = (r+s,0)$ and $(r,0)(s,0) = (rs,0)$. It is consequently a ring isomorphism onto the subring $R \times \{0\}$.